

RENAN MACÊDO

GAME DEVELOPER · REAL-TIME / MOCAP · VIRTUAL PRODUCTION

Unreal Engine · Unity · C# · Rokoko · Vive Mars · Aximmetry · Blueprints · Blender · Level Design | Jacareí, São Paulo, Brazil

[Portfolio](#) [LinkedIn](#) [GitHub](#) [ArtStation](#)

PROFILE

Game Developer and Real-Time Production Specialist with 10+ years of software and interactive media experience. Hands-on background in motion capture, virtual production and Unreal Engine, combined with game development, Level Design, gameplay programming, UI/UX and end-to-end product delivery. Experienced taking ideas from concept through implementation, testing, live operation and publication.

MOTION CAPTURE & VIRTUAL PRODUCTION

- **Motion capture:** ongoing hands-on production using Rokoko motion-capture suits, recording performances with a live performer for character work.
- **Virtual production:** owned end-to-end production workflows for live events and recorded commercial content, from capture and tracking to the final real-time scene.
- **Real-time systems:** Aximmetry, Vive Mars tracking, Unreal Engine Blueprints, live operation, scene setup and real-time troubleshooting.
- **Chroma key:** recorded and operated real-time Unreal environments for commercial productions using green-screen/chroma-key workflows.

GAME DEVELOPMENT

- **Game development:** Unreal Engine, Unity, C#, gameplay programming, game systems, RPG systems and UI/UX.
- **Game design:** mechanics, progression, player experience, gamification, turn-based systems and product thinking.
- **End-to-end development:** prototyping, implementation, integration, testing, publishing, iteration and live-product maintenance.
- **Technical / creative bridge:** comfortable moving between engine technology, visual production and game design requirements.

SELECTED IMPACT

- 20K+ downloads - Character Craft 5.5e
- 110K+ downloads - NTPlay streaming application
- 60K+ monthly active users - Radio NT
- 10+ years building software, games and interactive products
- Technical articles published by 80 Level on Unreal Engine procedural generation

PROFESSIONAL EXPERIENCE

REDE NOVO TEMPO DE COMUNICAÇÃO

Game Developer / Game Designer / Software Developer / Virtual Production

Brazil · Mar 2018 - Present

- Developed games and interactive applications, contributing to game design, gameplay programming, UI/UX, production coordination and launch support.
- Responsible for virtual production workflows spanning motion capture, camera/tracking systems, Aximmetry, Vive Mars and Unreal Engine for live events and recorded content.
- Created Unreal Engine Blueprints and operated real-time environments during productions, including chroma-key commercial shoots.
- Currently involved in ongoing Rokoko motion-capture recording with a performer for character production.

CODEBUDDY

Developer, Game Development Teacher & Educational Content Creator

Brazil · 2016 - 2018

- Taught programming, game development, game design, app development, robotics and gamification.
- Created educational and gamified learning content.

SELECTED GAME PROJECTS

A JORNADA

Primary Level Designer · Game Designer · Gameplay

Designed and implemented all levels of the game, including puzzle layouts, progression, mechanics and quests. Worked across level design, game design, gameplay systems, environmental interactions, player experience and UI/UX.

PULSE HERO

Fitness RPG · Current launch

Mobile RPG turning real-world fitness into character progression, combining progression systems, combat and gamification. Current focus includes game design, systems, Flutter development and product launch.

CHARACTER CRAFT 5.5E

20K+ downloads · RPG companion

Cross-platform RPG companion built with Flutter. Designed RPG data systems and online game rooms using WebSockets, Firebase, authentication, notifications, premium features and publishing.

PROCEDURAL ENVIRONMENTS

Unreal Engine · PCG

Procedural content workflows including a medieval village generator and UE5 PCG experiments, documented in technical articles.

TECHNICAL SKILLS

Real-Time: Unreal Engine, Blueprints, Aximmetry, Vive Mars, real-time production, motion capture

Game: Unity, C#, gameplay programming, game systems, Level Design, Game Design, UI/UX

3D / Visual: Blender, procedural generation, motion capture workflows, chroma key

Software: Flutter, Dart, Firebase, WebSockets, REST APIs, JSON

Platforms: Android, iOS, macOS, Windows

RENAN MACÊDO

GAME DEVELOPMENT · MOTION CAPTURE · VIRTUAL PRODUCTION

Selected technical background for game and real-time production roles

[Portfolio](#) [LinkedIn](#) [GitHub](#) [ArtStation](#)

MOTION CAPTURE / REAL-TIME PIPELINE

- **Capture:** Rokoko suit-based performance capture in ongoing character production.
- **Tracking:** Vive Mars for real-time camera / tracking workflows.
- **Engine integration:** Unreal Engine, Blueprints and real-time scene operation.
- **Virtual production:** Aximmetry-based live production for events and recorded commercial content.
- **Production formats:** live operation and chroma-key / green-screen recording.

PROFESSIONAL CONTEXT

- Worked across development, game design, visual production and product requirements.
- Comfortable working independently on technical setups and prototypes, as well as collaborating with multidisciplinary production teams.
- Experience troubleshooting real-time production issues under live and recorded conditions.

SOFTWARE & REAL-TIME SYSTEMS

- **Cross-platform:** Flutter, Dart, Android, iOS, macOS and Windows.
- **Backend & real-time:** WebSockets, Firebase, Firestore, REST APIs, JSON and synchronization.
- **Product delivery:** architecture, debugging, analytics, publishing and live maintenance.

EDUCATION & LANGUAGES

Technology Degree in Digital Games

Universidade FUMEC · 2013 - 2016

Portuguese: Native | **English:** Advanced

PUBLICATIONS

[80 Level - Procedural Medieval Village Generator Set-up in UE5](#)

[80 Level - Experimenting with UE5.2's Procedural Content Generation Framework](#)

LEVEL DESIGN & GAME DESIGN

- **A Jornada:** primary Level Designer; designed and implemented all game levels, puzzle layouts, progression, mechanics and quests.
- **Player experience:** worked on environmental interactions, grid movement, UI/UX and gameplay systems.
- **Systems thinking:** experience designing progression, RPG systems, mechanics and gamified loops.
- **End-to-end ownership:** comfortable taking a design idea through implementation, testing and iteration.

SELECTED PRODUCTS

NTPLAY & RADIO NT

60K+ MAU · 110K+ downloads

Video and audio streaming applications rebuilt in Flutter. Architecture, APIs, Firebase, authentication, notifications, UI/UX, analytics, publishing, debugging and ongoing maintenance.

CHARACTER CRAFT 5.5E

20K+ downloads

Cross-platform RPG companion combining RPG systems, real-time game rooms, authentication, premium features and publishing.

PULSE HERO

Fitness RPG · Current launch

Current product combining fitness, combat, progression, gamification, game design and cross-platform development.

BROADER TECHNICAL BACKGROUND

Game: Unity, Unreal Engine, C#, gameplay programming, RPG systems, progression, gamification, UI/UX

Content / Tools: Unreal Engine PCG, Blender, Git, Aximmetry, motion capture

Software: Flutter, Dart, Firebase, REST APIs, WebSockets, JSON